SCOREBOARD – Hadamard 14432 Paley-I (2026-10-09)

Cutoff: 2026-10-09T23:51:48Z (~7:51PM ET)

Object	Status	max|HHˆT-nI|	Wall (s)	SPK
H_14432 Paley-I (q=14431)	PASS	0	2436.1235	**IN SPK** (Paley-I)
H_14432 Paley-II (q=7215)	FAIL/SKIP	–	–	q=7215=3*5*13*37 NOT prime; ===3 mod 4 needs ===1
H_14432 Kronecker (4*3608)	SKIP deferred	–	–	3608===0 mod 4 order-admissible; H_3608 via Paley-I q=3607 ALSO ELIGIBLE – deferred

Units: 764/764 * row blocks 113/113 * overall_pass=True * wall_total=2474.4638s
Gap 2052: **OPEN** * clay_closed=false * GATE-04 **LOCKED** * conjecture **OPEN**
IN SPK via Paley-I; Miyamoto digression skipped; exclude Miyamoto-only 8948/8644/8516/...; prior SPK 10224+10212+10164+10160+10152+10140+10112+10104+10100+10068+10040+10008+9968+9932+9924+9908+9888+9884 retained

Next optional: **q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3ˆ2*71317 NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59ˆ2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2ˆ2*59ˆ2 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2ˆ2*3*1163 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3ˆ2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2ˆ2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7001->n=14004 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*1967 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3*1167 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2ˆ2*3*1167 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7013->n=14028 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7013 prime ===1 mod 4; Paley-I q=14027=13ˆ2*83 NOT prime -> FAIL; Kronecker 14028=4*3507; 3507=3*2389 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3507); 14028=2ˆ2*3*2389 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7057->n=14116 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7057 prime ===1 mod 4; Paley-I q=14115=3*5*941 NOT prime -> FAIL; Kronecker 14116=4*3529; 3529 odd prime (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3529);**

14116=2^2*3529 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7069->n=14140 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7069 prime ===1 mod 4; Paley-I q=14139=3^21571 NOT prime -> FAIL; Kronecker 14140=4*3535; 3535=5*7101 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3535); 14140=2^2*57101 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7109->n=14220 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7109 prime ===1 mod 4; Paley-I q=14219=59*241 NOT prime -> FAIL; Kronecker 14220=4*3555; 3555=3^2*79 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3555); 14220=2^2*3^2*79 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7129->n=14260 Pal
NOT prime -> FAIL; Kronecker 14260=4*3565; 3565=5*2331 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3565); 14260=2^2*5*2331 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | next Paley-I DIRECT after 14432 (and FIRST any direct route after 14432): q=14447->n=14448 (prime ===3 mod 4; only other prime in 14432..14446 is 14437 ===1 mod 4, skipped; Paley-II q=7223=31*233 NOT prime power (===3 mod 4) -> FAIL/SKIP; Kronecker 14448=4*3612; 3612=2^2*3*743 ===0 mod 4 order-admissible but H_3612 has NO direct classical Paley route (3611=23*157 composite; 1805=5*19^2 composite) -> FAIL/SKIP note-only (do NOT invent H_3612); 14448=2?*3*743 ===0 mod 16) | FIRST Paley-II DIRECT after 14432: q2=7229->n=14460 (q2=7229 prime ===1 mod 4; Paley-I q=14459=19*761 NOT prime -> FAIL; Kronecker 14460=4*3615; 3615=3*5*241 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3615); 14460=2^2*3*5*241 ===12 mod 16; needs a Paley-II wave (template hadamard_14428_paleyII / hadamard_14356_paleyII); comes AFTER the Paley-I 14448) | Kronecker-only (note-only): none in 14436..14448 (14448's H_3612 has no direct Paley route; 14436/14440/14444 m not ===0 mod 4) (or gap 2052 / Track B)